US 20250283486A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0283486 A1
(43) Pub. Date: Sep. 11, 2025
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(54) **DUST-COLLECTING FAN BLADE COVER DEVICE**

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(21) Appl. No.: **18/668,394**

(22) Filed: **May 20, 2024**

Related U.S. Application Data

(60) Provisional application No. 63/561,776, filed on Mar. 6, 2024.

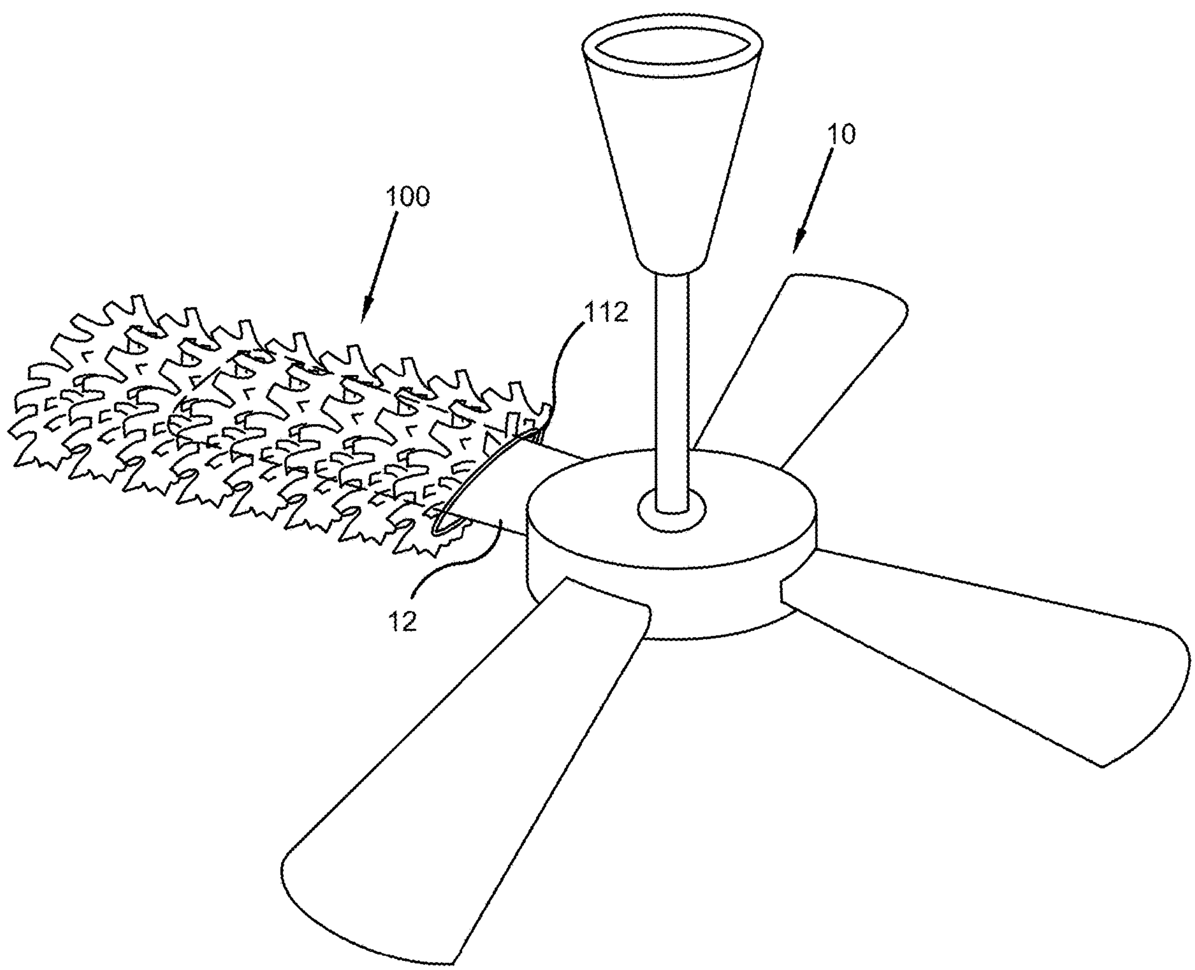
Publication Classification

(51) **Int. Cl.**
F04D 29/70 (2006.01)

(52) **U.S. Cl.**
CPC ***F04D 29/703*** (2013.01)

(57) **ABSTRACT**

A dust-collecting fan blade cover device is provided. The device is designed to be placed over a fan blade and comprises a body with at least one opening that may be equipped with a fastener to securely attach to the blade. The exterior surface of the body is designed to capture dust and debris from the air and can be made from materials like microfiber, cotton, or polyester, which may be treated to increase surface area for better dust collection. Additionally, this surface may be impregnated with a scent to release an aroma when the fan is in operation.



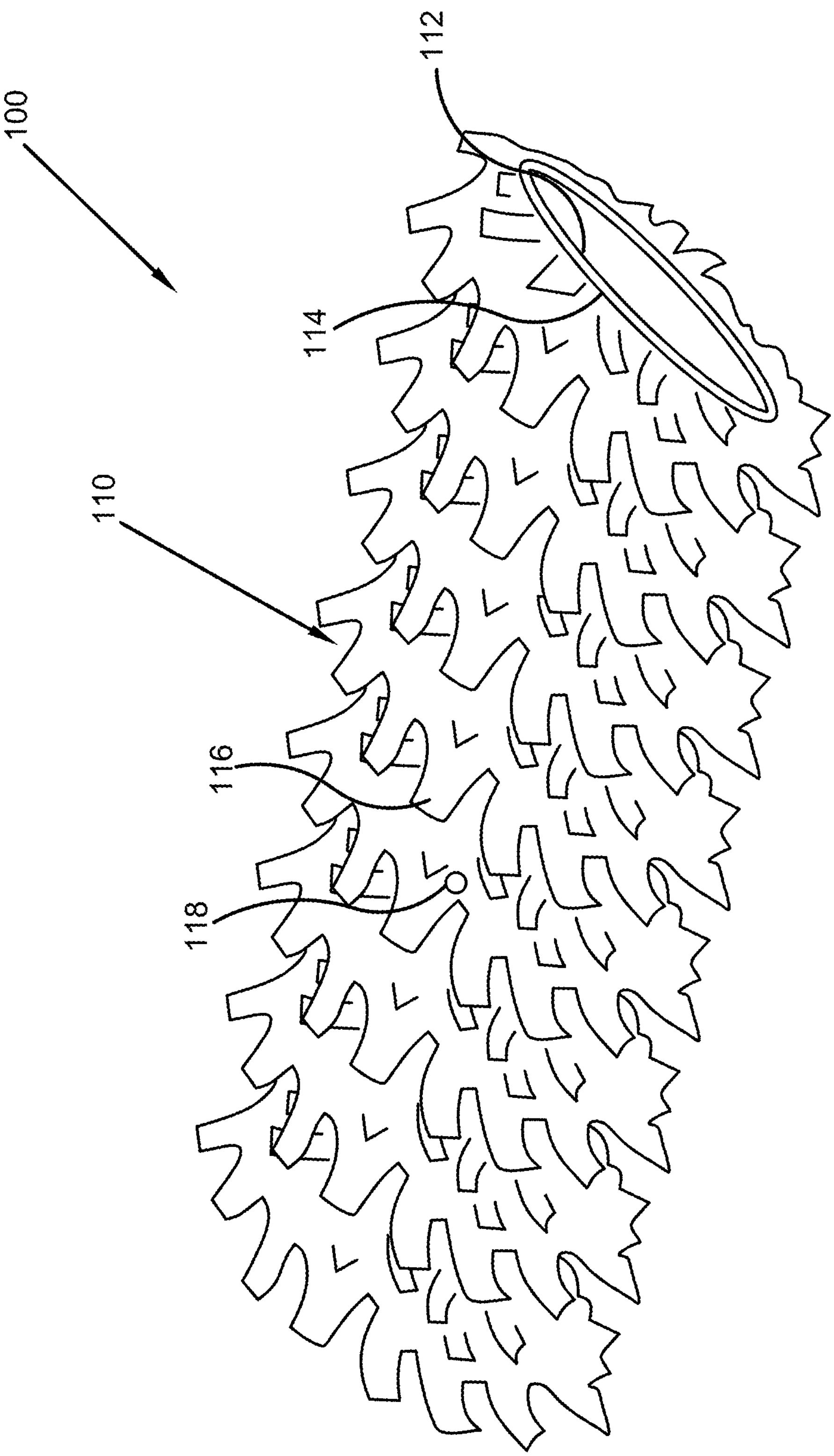


FIG. 1

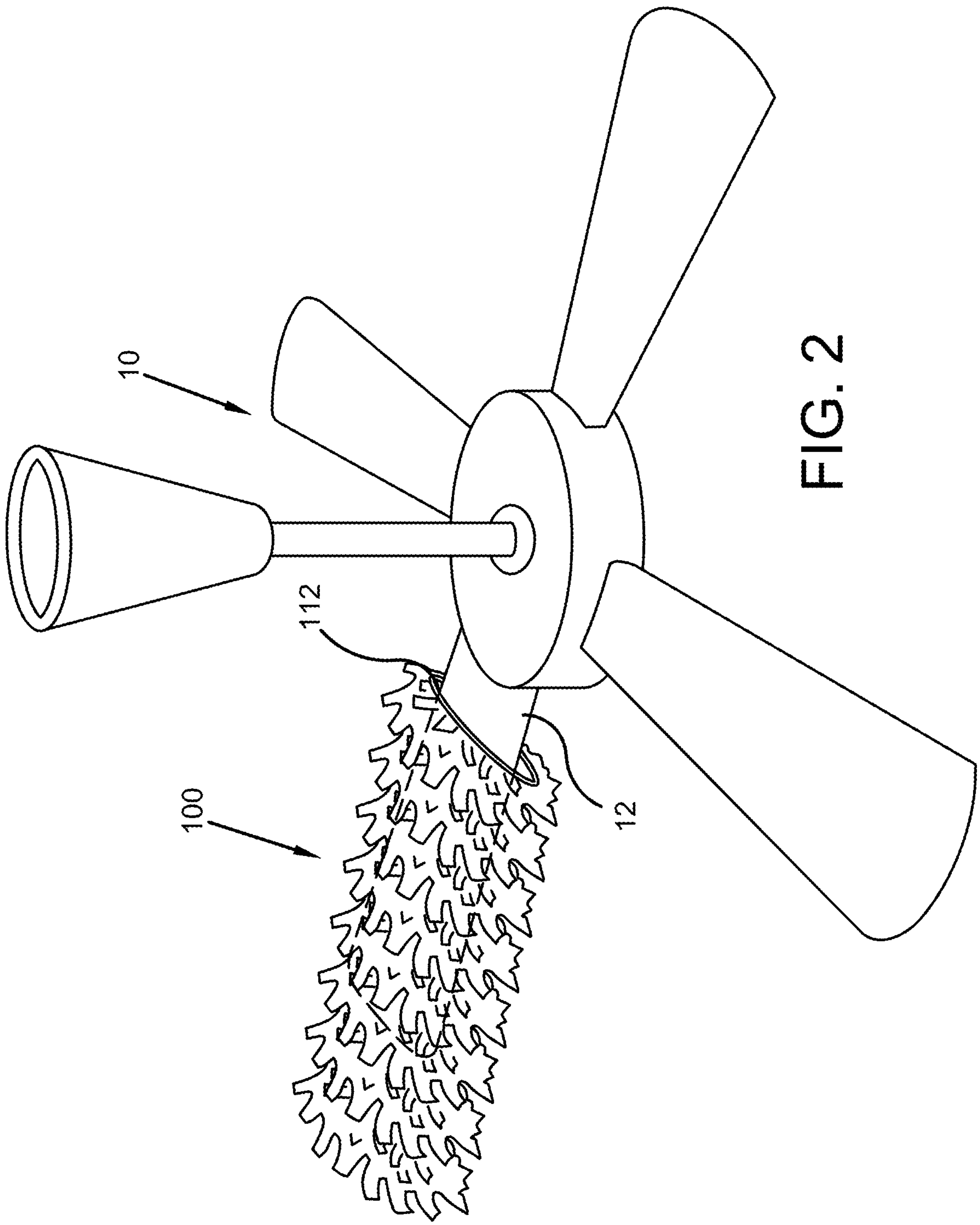


FIG. 2

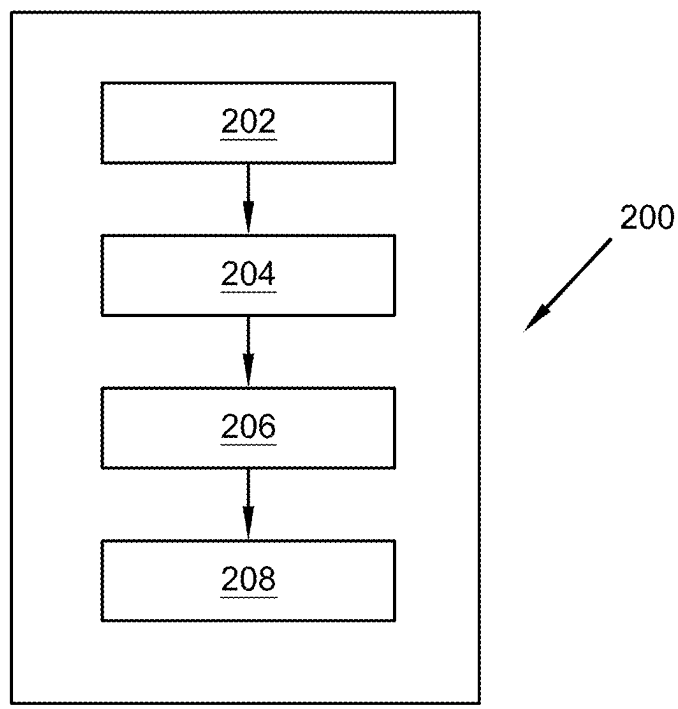


FIG. 3

DUST-COLLECTING FAN BLADE COVER DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to, and the benefit of, U.S. Provisional Application No. 63/561,776, which was filed on Mar. 6, 2024, and is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

[0002] The present invention relates generally to the field of dust collection devices. More specifically, the present invention relates to a dust-collecting fan blade cover device that can be placed over a fan blade, wherein the device captures dust and other particles while the fan blade spins. Accordingly, the present disclosure makes specific reference thereto. Nonetheless, it is to be appreciated that aspects of the present invention are also equally applicable to other like applications, devices, and methods of manufacture.

BACKGROUND

[0003] Maintaining cleanliness in various environments such as homes, offices, hospitals, and other public buildings presents significant challenges. Without regular attention, spaces can become cluttered and unsanitary. One common issue is that ceiling fans often gather considerable amounts of dust, which can go unnoticed until it becomes a larger problem. Additionally, many individuals and institutions may not have access to effective methods or tools to filter and remove airborne dust particles. This leads to frequent accumulation of dust and debris on various surfaces, necessitating more intensive cleaning efforts to maintain hygiene and comfort in these spaces.

[0004] Therefore, there exists a long-felt need in the art for an improved cleaning device. There also exists a long-felt need in the art for a dust-collecting fan blade cover device. More specifically, there exists a long-felt need in the art for a dust-collecting fan blade cover device that removes dust and other particles from the air. Further, there exists a long-felt need in the art for a dust-collecting fan blade cover device that removes dust and other particles from the air via a passive process.

[0005] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises a dust-collecting fan blade cover device. The device is designed to be placed over a fan blade and comprises a body with at least one opening that may be equipped with a fastener to securely attach to the blade. The exterior surface of the body is designed to capture dust and debris from the air and can be made from materials like microfiber, cotton, or polyester, which may be treated to increase surface area for better dust collection. Additionally, this surface may be impregnated with a scent to release an aroma when the fan is in operation.

[0006] In this manner, the dust-collecting fan blade cover device of the present invention accomplishes all the forgoing objectives and provides a device for a dust-collecting fan blade cover device. More specifically, the device removes dust and other particles from the air. Further, there exists a long-felt need in the art for a dust-collecting fan blade cover device that removes dust and other particles from the air via a passive process while a fan spins.

SUMMARY

[0007] The following presents a simplified summary to provide a basic understanding of some aspects of the disclosed innovation. This summary is not an extensive overview, and it is not intended to identify key/critical elements or to delineate the scope thereof. Its sole purpose is to present some general concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0008] The subject matter disclosed and claimed herein, in one embodiment thereof, comprises a dust-collecting fan blade cover device comprised of a body that can be placed over the blade of a fan. As a result, the exterior surface of the body collects dust/debris circulating within the air of a room as the fan spins.

[0009] The body is preferably made from a fabric material and is preferably generally rectangular in shape (to fit over a fan blade). The body is comprised of at least one opening that allows the body to be placed over and surround a fan blade.

[0010] The exterior surface of the body captures dust and other debris in the air. To do so, the exterior surface may be made from a plurality of materials. Said material may be bristled, feathered, or otherwise separated in order to create more surface area for catching dust.

[0011] The present invention is also comprised of a method of using the device. First, a device is provided comprised of a body comprised of at least one opening. Then, the body can be placed over a fan blade, and may be optionally secured to/around the blade via at least one fastener. Next, the fan can be turned on such that the body collects dust and other particles out of the air as the blade spins. The body can then be removed from the blade and disposed of and/or washed/cleaned after a sufficient amount of dust/particles have been collected.

[0012] Accordingly, the dust-collecting fan blade cover device of the present invention is particularly advantageous as it provides a device for a dust-collecting fan blade cover device. More specifically, the device removes dust and other particles from the air. Further, there exists a long-felt need in the art for a dust-collecting fan blade cover device that removes dust and other particles from the air via a passive process while a fan spins. In this manner, the dust-collecting fan blade cover device overcomes the limitations of existing methods and devices used for dust cleaning known in the art.

[0013] To the accomplishment of the foregoing and related ends, certain illustrative aspects of the disclosed innovation are described herein in connection with the following description and the annexed drawings. These aspects are indicative, however, of but a few of the various ways in which the principles disclosed herein can be employed and are intended to include all such aspects and their equivalents. Other advantages and novel features will become apparent from the following detailed description when considered in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The description refers to provided drawings in which similar reference characters refer to similar parts throughout the different views, and in which:

[0015] FIG. 1 illustrates a perspective view of one potential embodiment of a dust-collecting fan blade cover device of the present invention in accordance with the disclosed architecture;

[0016] FIG. 2 illustrates a perspective view of one potential embodiment of a dust-collecting fan blade cover device of the present invention while attached around a fan blade in accordance with the disclosed architecture; and

[0017] FIG. 3 illustrates a flowchart of a method of using one potential embodiment of a dust-collecting fan blade cover device of the present invention in accordance with the disclosed architecture.

DETAILED DESCRIPTION

[0018] The innovation is now described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. In the following description, for purposes of explanation, numerous specific details are set forth to provide a thorough understanding thereof. It may be evident, however, that the innovation can be practiced without these specific details. In other instances, well-known structures and devices are shown in block diagram form to facilitate a description thereof. Various embodiments are discussed hereinafter. It should be noted that the figures are described only to facilitate the description of the embodiments. They are not intended as an exhaustive description of the invention and do not limit the scope of the invention. Additionally, an illustrated embodiment need not have all the aspects or advantages shown. Thus, in other embodiments, any of the features described herein from different embodiments may be combined.

[0019] As noted above, there exists a long-felt need in the art for an improved cleaning device. There also exists a long-felt need in the art for a dust-collecting fan blade cover device. More specifically, there exists a long-felt need in the art for a dust-collecting fan blade cover device that removes dust and other particles from the air. Further, there exists a long-felt need in the art for a dust-collecting fan blade cover device that removes dust and other particles from the air via a passive process.

[0020] The present invention, in one exemplary embodiment, is comprised of a dust-collecting fan blade cover device. The device is comprised of a body that can be placed over the blade of a fan such that the exterior surface of the body collects dust/debris circulating within the air of a room as the fan spins.

[0021] The body is preferably made from a fabric material and is preferably generally rectangular in shape (to fit over a fan blade). The body is comprised of at least one opening that allows the body to be placed over and surround a fan blade.

[0022] The exterior surface of the body captures dust and other debris in the air. To do so, the exterior surface may be made from a plurality of materials. Said material may be bristled, feathered, or otherwise separated in order to create more surface area for catching dust.

[0023] The present invention is also comprised of a method of using the device. First, a device is provided comprised of a body comprised of at least one opening. Then, the body can be placed over a fan blade, and may be optionally secured to/around the blade via at least one fastener. Next, the fan can be turned on such that the body collects dust and other particles out of the air as the blade spins. The body can then be removed from the blade and disposed of and/or washed/cleaned after a sufficient amount of dust/particles have been collected.

[0024] Accordingly, the dust-collecting fan blade cover device of the present invention is particularly advantageous as it provides a device for a dust-collecting fan blade cover device. More specifically, the device removes dust and other particles from the air. Further, there exists a long-felt need in the art for a dust-collecting fan blade cover device that removes dust and other particles from the air via a passive process while a fan spins. In this manner, the dust-collecting fan blade cover device overcomes the limitations of existing methods and devices used for dust cleaning known in the art.

[0025] Referring initially to the drawings, FIG. 1 illustrates a perspective view of one potential embodiment of a dust-collecting fan blade cover device **100** of the present invention in accordance with the disclosed architecture. The device **100** is comprised of a body **110** that can be placed over the blade **12** of a fan **10**. As a result, the exterior surface **116** of the body **110** collects dust/debris circulating within the air of a room as the fan **10** spins.

[0026] The body **110** is preferably made from a fabric material. However, the body **110** may be made from any material. The body **110** is preferably generally rectangular in shape (to fit over a fan blade **12**) but may be any shape. The body **110** is comprised of at least one opening **112** that allows the body **110** to be placed over and surround a fan blade **12**, as seen in FIG. **2**. The opening **112** may be comprised of at least one fastener **114** that secures the opening **112** around and/or to the fan blade **12**. Said fastener includes but is not limited to an elastic fastener, a hook and loop fastener, an adhesive fastener, a magnet fastener, etc.

[0027] The exterior surface **116** of the body **110** captures dust and other debris in the air. To do so, the exterior surface **116** may be made from a plurality of materials. Said materials include but are not limited to a microfiber, a cotton, an electrostatic cloth, a lambswool, a feather, a polyester, a chenille, etc. wherein said surface **116** and material may be bristled, feathered, or otherwise separated in order to create more surface area for catching dust.

[0028] In one embodiment, the exterior surface **116** is impregnated with at least one scent **118**. The scent **118** may be any natural or synthetic scent. As a result, when the device **100** is placed over a fan blade **12** and the fan **10** is powered on such that the device **100** catches dust, the device **100** also emits a pleasant scent.

[0029] The present invention is also comprised of a method of using **200** the device **100**, as seen in FIG. **3**. First, a device **100** is provided comprised of a body **110** comprised of at least one opening **112** [Step **202**]. Then, the body **110** can be placed over a fan blade **12**, and may be optionally secured to/around the blade **12** via at least one fastener **114** [Step **204**]. Next, the fan **10** can be turned on such that the body **110** collects dust and other particles out of the air as the blade **12** spins [Step **206**]. The body **110** can then be removed from the blade **112** and disposed of and/or washed/cleaned after a sufficient amount of dust/particles have been collected [Step **208**].

[0030] Certain terms are used throughout the following description and claims to refer to particular features or components. As one skilled in the art will appreciate, different persons may refer to the same feature or component by different names. This document does not intend to distinguish between components or features that differ in name but not structure or function. As used herein "dust-collecting fan blade cover device" and "device" are interchangeable and refer to the dust-collecting fan blade cover device **100** of the present invention.

[0031] Notwithstanding the forgoing, the dust-collecting fan blade cover device **100** of the present invention and its various components can be of any suitable size and configuration as is known in the art without affecting the overall concept of the invention, provided that they accomplish the above-stated objectives. One of ordinary skill in the art will appreciate that the size, configuration, and material of the dust-collecting fan blade cover device **100** as shown in the FIGS. are for illustrative purposes only, and that many other sizes and shapes of the dust-collecting fan blade cover device **100** are well within the scope of the present disclosure. Although the dimensions of the dust-collecting fan blade cover device **100** are important design parameters for user convenience, the dust-collecting fan blade cover device **100** may be of any size, shape, and/or configuration that ensures optimal performance during use and/or that suits the user's needs and/or preferences.

[0032] Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. While the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

[0033] What has been described above includes examples of the claimed subject matter. It is, of course, not possible to describe every conceivable combination of components or methodologies for purposes of describing the claimed subject matter, but one of ordinary skill in the art may recognize that many further combinations and permutations of the claimed subject matter are possible. Accordingly, the claimed subject matter is intended to embrace all such alterations, modifications, and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term "includes" is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term "comprising" as "comprising" is interpreted when employed as a transitional word in a claim.

1. A dust-collecting fan blade cover device comprising:

a body comprised of an exterior surface and an opening comprising a hook and loop fastener; and

wherein the exterior surface is comprised of a lambswool; and

wherein the exterior surface is impregnated with a natural scent.

2. The dust-collecting fan blade cover device of claim **1**, wherein the body is comprised of a fabric material.

3. The dust-collecting fan blade cover device of claim **1**, wherein the body is comprised of a rectangular shape.

4. (canceled)

5. (canceled)

6. The dust-collecting fan blade cover device of claim **1**, wherein the exterior surface is comprised of a bristled surface.

7. (canceled)

8. A dust-collecting fan blade cover device comprising:

a body comprised of an exterior surface, and an opening, the opening comprised of an adhesive fastener; and

wherein the exterior surface is comprised of a chenille and is feathered; and

wherein the exterior surface is impregnated with a synthetic scent.

9. (canceled)

10. (canceled)

11. (canceled)

12. The dust-collecting fan blade cover device of claim **8**, wherein the body is comprised of a fabric material.

13. The dust-collecting fan blade cover device of claim **8**, wherein the body is comprised of a rectangular shape.

14. (canceled)

15. A method of using a dust-collecting fan blade cover device, the method comprising the following steps:

providing a dust-collecting fan blade cover device comprised of a body comprised of an opening;

placing the body over a fan blade of a fan;

magnetically securing the opening around the fan blade;

turning on the fan; and

removing the body from the fan blade; and

wherein the body is comprised of a polyester exterior surface.

16. (canceled)

17. (canceled)

18. (canceled)

19. The method of using a dust-collecting fan blade cover device of claim **15**, wherein the body is comprised of a fabric material.

20. The method of using a dust-collecting fan blade cover device of claim **15**, wherein the exterior surface is comprised of a bristle.

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